



Subashini M 251801338-2025-AIDS ▾

S2

Started on	Thursday, 24 September 2026, 8:20 AM
State	Finished
Completed on	Thursday, 24 September 2026, 8:36 AM
Time taken	16 mins 19 secs
Marks	1.00/1.00
Grade	10.00 out of 10.00 (100%)

Question 1 | Correct Mark 1.00 out of 1.00

Write a Program to Implement the Quick Sort Algorithm

Input Format:

The first line contains the no of elements in the list-n

The next n lines contain the elements.

Output:

Sorted list of elements

For example:

Input	Result
5 67 34 12 98 78	12 34 67 78 98

Answer:

```

1  #include <stdio.h>
2
3  void swap(int* a, int* b) {
4      int t = *a;
5      *a = *b;
6      *a = *b;
7      *b = t;
8  }
9
10 int partition(int arr[], int low, int high) {
11     int pivot = arr[high];
12     int i = (low - 1);
13
14     for (int j = low; j <= high - 1; j++) {
15         if (arr[j] <= pivot) {
16             i++;
17             swap(&arr[i], &arr[j]);
18         }
19     }
20     swap(&arr[i + 1], &arr[high]);
21     return (i + 1);
22 }
23
24 void quickSort(int arr[], int low, int high) {
25     if (low < high) {
26
27         int pi = partition(arr, low, high);
28
29         quickSort(arr, low, pi - 1);
30         quickSort(arr, pi + 1, high);
31     }
32 }
33
34 int main() {
35     int n;
36
37     if (scanf("%d", &n) != 1) {
38         return 1;
39     }
40
41     int arr[n];
42
43     for (int i = 0; i < n; i++) {
44         if (scanf("%d", &arr[i]) != 1) {
45             return 1;
46         }
47     }
48
49     quickSort(arr, 0, n - 1);
50
51     for (int i = 0; i < n; i++) {
52         printf("%d ", arr[i]);

```

	Input	Expected	Got	
✓	5 67 34 12 98 78	12 34 67 78 98	12 34 67 78 98	✓
✓	10 1 56 78 90 32 56 11 10 90 114	1 10 11 32 56 56 78 90 90 114	1 10 11 32 56 56 78 90 90 114	✓
✓	12 9 8 7 6 5 4 3 2 1 10 11 90	1 2 3 4 5 6 7 8 9 10 11 90	1 2 3 4 5 6 7 8 9 10 11 90	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

[Back to Course](#)